

# An amplitude-level model of $\pi^0$ and $\eta$ electroproduction fitted to CLAS, Hall A and COMPASS

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## Abstract

A model of exclusive  $\pi^0$  and  $\eta$  electroproduction is built directly at the level of the helicity amplitudes, without reference to generalised parton distributions, and fitted to 1555 measured points from CLAS6, CLAS12, Hall A and COMPASS. The purpose is not to test a particular GPD parametrisation but to establish what the data themselves demand of the amplitudes, so that any model can be measured against it. The fit has 28 free parameters and gives  $\chi^2/\text{ndf} = 1.3918$  (2125.2/1527). Four conclusions survive a full leave-one-out analysis: the flavour ratio  $n/p$  is separated by four neutron points and by nothing else; one polarised-target experiment alone fixes the relative phase of two transverse amplitudes;  $\sigma_{LT}$  is the model's systematic failure, seen independently in three data sets; and a single power  $Q^{n_Q}$  cannot span  $Q^2$  from 1.2 to 8.3 GeV<sup>2</sup>.

## 1 The model

### 1.1 Amplitudes

The observables are built from six channel helicity amplitudes. In the notation of the  $\pi^0$  amplitude note,

|                |                                |                |                            |
|----------------|--------------------------------|----------------|----------------------------|
| $T_{00}^{(+)}$ | longitudinal, recoil non-flip  | $T_{00}^{(-)}$ | longitudinal, recoil flip  |
| $T_{01}^{(+)}$ | natural transverse, non-flip   | $T_{01}^{(-)}$ | natural transverse, flip   |
| $U_{01}^{(+)}$ | unnatural transverse, non-flip | $U_{01}^{(-)}$ | unnatural transverse, flip |

At the level used here  $T_{01}^{(-)} = U_{01}^{(-)} = D/2$ , real and equal, which is the forward-kinematics relation; the remaining four carry the fitted moduli and phases.

The amplitudes are constructed from flavour generalised form factors. Each block  $X \in \{H_T^u, H_T^d, \bar{E}_T^u, \bar{E}_T^d, L\}$  has the form

$$X(t, x_B, Q^2) = N \exp[(b + b' \ln x_B) t] (Q^2)^{n_Q/2}, \quad (1)$$

so that  $b$  is the  $t$ -slope extrapolated to  $x_B = 1$  and  $-b'$  plays the role of the Regge  $\alpha'$ . This is the same convention as the Goloskokov–Kroll fits. Note that an older parameter convention used  $L = \ln x_B - \ln 0.15$ ; files in that convention must be converted, never copied raw.

The flavour combinations entering each channel are

$$\pi^0 p: \frac{2X_u + X_d}{3\sqrt{2}}, \quad \eta p: \frac{2X_u - X_d}{3\sqrt{6}k}, \quad \pi^0 n: \frac{X_u + 2X_d}{3\sqrt{2}}, \quad (2)$$

with  $k = 0.863$  in the transverse twist-3 sector (chiral condensate factors) and  $k = 0.695$  in the longitudinal twist-2 sector (decay constants only). The near cancellation  $2X_u - X_d$  in the  $\eta$  channel is what makes that channel sensitive to the flavour separation, and it is the reason  $\eta$  data carry weight far beyond their number.

Writing  $\xi = \frac{x_B}{2-x_B} (1 + M_p^2/Q^2)$ ,  $t' = t - t_{\min}$  and  $A = A(Q^2, x_B)$  for the phase-space factor,

$$D = \sqrt{2A(1-\xi^2)} \langle H_T \rangle, \quad U_{01}^{(+)} = \sqrt{A\kappa} \langle \bar{E}_T \rangle, \quad |T_{00}^{(-)}| = \sqrt{A\kappa} \langle L \rangle, \quad (3)$$

with  $\kappa = -t'/(8M_p^2)$  and  $\langle \cdot \rangle$  the flavour combination above. Four further parameters unlock what would otherwise be exact relations:  $\rho_{CE}$  and  $\phi_{CE}$  give  $T_{01}^{(+)}$  a modulus and phase relative to  $U_{01}^{(+)}$  (breaking  $C = E$ ),  $\phi_w$  is the phase of  $U_{01}^{(+)}$ , and  $\rho_{nf}, \phi_{nf}$  give the longitudinal non-flip amplitude  $T_{00}^{(+)}$ , which vanishes at leading order. The phase of  $T_{00}^{(-)}$  is  $\delta_0 + \delta_1 t$ , common to both flavours.

Finally, each  $d$ -flavour convolution carries a phase relative to  $u$ . The overall phase of a channel is unobservable, so  $u$  is kept real and only  $d$  carries  $\varphi_{d/u}$ , separately for  $H_T, \bar{E}_T$  and  $L$ . These are physical: the real and imaginary parts of a convolution have different  $t$ -slopes, so  $u$  and  $d$  cannot share a phase, and the  $\eta$  cancellation therefore ceases to be exact.

## 1.2 From amplitudes to structure functions

The nine structure functions follow from the Fourier dictionary, verified in the note against the QED photon density matrix:

$$\begin{aligned} \sigma_L &= \sum |T_{00}|^2, & \sigma_T &= \sum |T_{01}|^2 + \sum |U_{01}|^2, & \sigma_{TT} &= \sum |T_{01}|^2 - \sum |U_{01}|^2, \\ \sigma_{LT} + i\sigma_{LT'} &= -\sqrt{2} \sum T_{00}^* T_{01}, & \sigma_{LT'}^{LL} + i\sigma_{LT}^{UL} &= -\sqrt{2} \sum T_{00}^* U_{01}, & \sigma_{T'} + i\sigma_{TT}^{UL} &= 2 \sum T_{01}^* U_{01}. \end{aligned} \quad (4)$$

Only the moduli of  $T_{01}$  and  $U_{01}$  enter  $\sigma_T$  and  $\sigma_{TT}$ ; their *relative phase* appears only in  $\sigma_{T'}$ . This observation matters later: it is why one polarised-target experiment is the sole constraint on that phase.

The measured cross section is the standard decomposition

$$\frac{d^2\sigma}{dt d\phi} = \frac{1}{2\pi} \left[ \sigma_T + \epsilon \sigma_L + \epsilon \cos 2\phi \sigma_{TT} + \sqrt{2\epsilon(1+\epsilon)} \cos \phi \sigma_{LT} + h \sqrt{2\epsilon(1-\epsilon)} \sin \phi \sigma_{LT'} \right], \quad (5)$$

identical to eq. (B4) of the CLAS6 paper and eq. (2) of COMPASS, with  $\phi$  in the Trento convention and  $h$  the beam helicity.

## 2 The data

Thirteen data sets are loaded from a single workbook, `pi0_eta_database.xlsx`. Nothing enters the fit from any other source: a second source is how an old, mis-weighted version of one set once survived unnoticed. Table 1 lists everything, fitted or not.

Table 1: The database and the fit. A set marked  $--$  is not fitted; its  $\chi^2$  is then a blind prediction. CLAS12 cross sections are preliminary; the published De Masi  $\alpha$  are the same measurement as the  $\phi$  distributions and would be double counting.

| set                            | key            | fitted | points | $\chi^2$ | per point |
|--------------------------------|----------------|--------|--------|----------|-----------|
| CLAS6 $\pi^0$ p                | clas6_pi0      | yes    | 288    | 484.2    | 1.68      |
| CLAS6 $\eta$ p                 | clas6_eta      | yes    | 207    | 216.3    | 1.04      |
| Hall A 2017, $\pi^0$ n         | halla_n        | yes    | 12     | 20.7     | 1.72      |
| Hall A 2016, $\pi^0$ p         | halla_y16      | yes    | 21     | 94.0     | 4.47      |
| Hall A 2011, $\pi^0$ p         | halla_y11      | yes    | 112    | 146.3    | 1.31      |
| Hall A 2021, $\pi^0$ p         | halla_y21      | yes    | 144    | 338.2    | 2.35      |
| CLAS12 cross sections          | clas12_xs      | --     | 438    | 3830.9   | 8.75      |
| COMPASS 2025                   | compass        | yes    | 26     | 64.9     | 2.49      |
| De Masi, published $\alpha$    | bsa_demasi     | --     | 56     | 81.7     | 1.46      |
| Zhao, $\eta$ BSA               | bsa_zhao       | yes    | 12     | 13.6     | 1.13      |
| CLAS12 $\sigma_{LT'}/\sigma_0$ | bsa_clas12     | yes    | 30     | 46.1     | 1.54      |
| De Masi, $A_{LU}(\phi)$        | bsa_demasi_phi | yes    | 663    | 620.7    | 0.94      |
| eg1-dvcs, pol. target          | eg1            | yes    | 40     | 73.9     | 1.85      |

## 2.1 Provenance of the uncertainties

Every weight in the fit now comes from a publication. Three defects were repaired on 5 September 2026 and are recorded here because they changed the answer, not merely the bookkeeping.

**De Masi carried no systematic at all.** The CLAS physics database export gives statistical errors only, so 703  $\phi$  points were the most heavily weighted set in the sample. Ref. PRC **77** 042201(R), p. 3, quotes 0.016 from event selection and from the choice of fit function; it is now applied to all 703 points and to the 62 published moments.

**The Hall A neutron systematics exist only as a picture.** PRL **118** 222002 gives no table; the point-to-point uncertainties appear only as the shaded bands of its figure 5. They were read from the vector PDF, with the vertical calibration taken not from the axis but from the published error bars themselves — half a bar in figure units against the quoted statistical error. The calibration closes to 0.0 % in the  $\sigma_T$  panel and 1.8 % in  $\sigma_{LT}$  and  $\sigma_{TT}$ . The bands are asymmetric, in places by a factor three; they are symmetrised as (up + down)/2, and the 3.1 % normalisation they contain is removed in quadrature because it is fitted separately.

**An invented 10 % was removed.** Hall A 2016 and the neutron had been carrying a flat 10 % of the author’s own making. It was wrong in both directions at once: it concealed a real discrepancy in the 2016 proton set (3.93  $\rightarrow$  5.78 per point once removed) and inflated a false one in the neutron, whose true bands are larger than 10 %. An invented uncertainty is worse than a missing one, because it looks like knowledge.

## 2.2 Normalisation uncertainties

Three sets carry an overall normalisation that multiplies every point together: 3.5 % for De Masi (beam polarisation), 3.12 % for Hall A 2016 (ref. PRL **117** 262001, table III) and 3.1 % for the neutron. Adding these in quadrature to the per-point errors would discard the correlation and smear each set instead of letting it move as a whole. They are therefore fitted as one free scale per set,

$$r = \frac{\text{data} - \lambda \text{model}}{\sigma}, \quad \text{with a penalty} \quad \left( \frac{\lambda - 1}{\sigma_{\text{norm}}} \right)^2 \text{ counted in } \chi^2, \quad (6)$$

and the number of degrees of freedom reduced by the number of  $\lambda$ ; otherwise a nuisance would be a free way to improve  $\chi^2/\text{ndf}$ . Table 2 gives the results. Hall A 2016 is the one to notice: its normalisation is dragged nearly two standard deviations and the set still does not fit.

Table 2: Fitted normalisations.

| set            | prior  | $\lambda$           | pull           |
|----------------|--------|---------------------|----------------|
| bsa_demasi_phi | 3.50 % | $1.0587 \pm 0.0326$ | $+1.68 \sigma$ |
| halla_n        | 3.10 % | $0.9776 \pm 0.0304$ | $-0.72 \sigma$ |
| halla_y16      | 3.12 % | $1.0562 \pm 0.0253$ | $+1.80 \sigma$ |

## 3 The fit

### 3.1 Constraints

Two constraints are imposed for stated physical reasons rather than to improve  $\chi^2$ .

$n_{Q,d} = n_{Q,u}$  **in both transverse sectors.** A generalised form factor is (GPD)  $\times$  (hard subprocess). The GPD carries no  $Q^2$  dependence, and the subprocess is flavour-blind up to the quark charge. Transversity does not mix with gluons, so its evolution is non-singlet and flavour-blind as well. Left free, the fitted  $n_{Q,d} - n_{Q,u}$  came out  $-7.0$  and  $+2.8$  in successive fits, with flavour ratios running by an order of magnitude across  $Q^2$ .

$b_d = b_u$  **for  $H_T$ .** Left free, the fit drives  $b_d$  to whatever ceiling it is given — 12, 18, 25 — buying two units of  $\chi^2$  while the normalisation  $N_d$  compensates by a factor sixteen. It is a flat valley, not a measurement.

In addition every  $t$ -slope is required non-negative over  $x_B \in [0.05, 1]$ , not only where there are data: without it  $H_T^u$  came out with  $b = -0.26$ , so that the form factor *grew* with  $|t|$  above  $x_B = 0.785$ .

### 3.2 Parameters

Tables 3 and 4 give all 28 fitted parameters with errors from the Jacobian at the minimum. A dagger marks a parameter sitting on a bound, where the quoted error is one-sided and the value is not a measurement.

Table 3: Block parameters, eq. (1).  $N$  in  $\text{nb}^{1/2}/\text{GeV}$ ,  $b$  and  $b'$  in  $\text{GeV}^{-2}$ ,  $n_Q$  dimensionless.

| block         | $N$                  | $b$                       | $b'$                | $n_Q$                  |
|---------------|----------------------|---------------------------|---------------------|------------------------|
| $H_T^u$       | $20.633 \pm 0.626$   | $0.000 \pm 0.043^\dagger$ | $-1.086 \pm 0.061$  | $1.227 \pm 0.031$      |
| $H_T^d$       | $-6.480 \pm 5.490$   | $= b_u$                   | $= b'_u$            | $= n_{Q,u}$            |
| $\bar{E}_T^u$ | $60.785 \pm 2.884$   | $0.678 \pm 0.049$         | $-0.312 \pm 0.036$  | $1.164 \pm 0.065$      |
| $\bar{E}_T^d$ | $229.131 \pm 21.224$ | $5.366 \pm 0.358$         | $= b'_u(\bar{E}_T)$ | $= n_{Q,u}(\bar{E}_T)$ |
| $L$ (T00)     | $36.118 \pm 4.442$   | $0.000 \pm 0.043^\dagger$ | $-1.269 \pm 0.104$  | $0.390 \pm 0.144$      |

Table 4: Remaining parameters, with the interval each was allowed.

| parameter                  | value              | lower | upper |
|----------------------------|--------------------|-------|-------|
| $R_L = N_d/N_u$ (T00)      | $-0.652 \pm 0.177$ | -2.00 | 2.00  |
| $\delta_0$                 | $1.727 \pm 0.082$  | -3.14 | 3.14  |
| $\delta_1$                 | $0.695 \pm 0.088$  | -3.00 | 3.00  |
| $\rho_{CE}$                | $0.170 \pm 0.021$  | 0.00  | 1.50  |
| $\phi_{CE}$ (frozen)       | $0.000 \pm 0.000$  | -3.14 | 3.14  |
| $\phi_w$                   | $-0.586 \pm 0.125$ | -3.14 | 3.14  |
| $\rho_{nf}$                | $1.726 \pm 0.201$  | 0.00  | 3.00  |
| $\phi_{nf}$                | $2.107 \pm 0.148$  | -3.14 | 3.14  |
| $\varphi_{d/u}(H_T)$       | $-1.254 \pm 0.277$ | -3.14 | 3.14  |
| $\varphi_{d/u}(\bar{E}_T)$ | $0.179 \pm 0.451$  | -3.14 | 3.14  |
| $\varphi_{d/u}(T_{00})$    | $0.502 \pm 0.337$  | -3.14 | 3.14  |

Two features deserve comment.  $N_d$  of  $H_T$  is consistent with zero: the  $d$ -quark transversity normalisation is not determined by these data at all. And the  $d/u$  phase of  $\bar{E}_T$  is likewise consistent with zero, while only the  $H_T$  phase differs from zero, by about  $4.5\sigma$ .

### 3.3 Derived quantities

At  $Q^2 = 1.75 \text{ GeV}^2$ ,  $x_B = 0.36$ ,  $-t = 0.27 \text{ GeV}^2$ :

$$\left. \frac{d}{u} \right|_{H_T} = -0.314, \quad \left. \frac{d}{u} \right|_{\bar{E}_T} = 0.924, \quad \left. \frac{n}{p} \right|_{\sigma_{TT}} = 1.042, \quad \frac{\sigma_L}{\sigma_T} = 0.080.$$

The  $n/p$  ratio carries a caveat that is not a formality; see section 5.

## 4 What the leave-one-out analysis shows

Each data set was removed in turn and the fit repeated, in two series: without and with Hall A 2021. A set's  $\chi^2$  when it is *not* fitted is a blind prediction, and is the useful number: a set that is left out and still described brings no unique information.

Table 5: Blind  $\chi^2$  per point of the set that was removed, and the effect on two derived quantities. Series 8 has 9 sets, series B adds Hall A 2021.

| removed        | blind, series 8 | blind, series B | $d/u _{H_T}$ | $n/p$        |
|----------------|-----------------|-----------------|--------------|--------------|
| — (reference)  | —               | —               | −0.311       | 1.062        |
| CLAS6 $\pi^0$  | 2.89            | 3.52            | −0.322       | 0.973        |
| CLAS6 $\eta$   | <b>218.9</b>    | <b>2.81</b>     | −0.475       | 1.220        |
| Hall A neutron | 10.06           | <b>13.06</b>    | +0.401       | <b>2.153</b> |
| Hall A 2016    | 6.18            | 5.90            | −0.335       | 1.039        |
| Hall A 2011    | 2.66            | 1.93            | −0.354       | 1.094        |
| Hall A 2021    | —               | 13.10           | −0.238       | 1.037        |
| De Masi $\phi$ | 1.02            | 1.02            | −0.277       | 1.072        |
| Zhao           | 2.25            | 1.90            | −0.237       | 1.086        |
| CLAS12 BSA     | 2.88            | 2.59            | −0.334       | 1.044        |
| eg1-dvcs       | <b>40.16</b>    | <b>41.48</b>    | −0.405       | 0.648        |

**Hall A 2021 has taken the flavour separation over from  $\eta$ .** Without it, removing the  $\eta$  data left the model predicting  $\eta$  at 219 per point. With it in the fit that collapses to 2.81. Hall A 2021 reaches  $x_B = 0.60$ , the highest in the database, and now covers the region that only the  $\eta$  cancellation used to probe.  $\eta$  still moves  $d/u|_{H_T}$  from −0.311 to −0.475 when dropped; it is no longer irreplaceable.

**The reference fit is reproducible.** Removing Hall A 2021 from series B reproduces the series 8 reference to four figures, from a different starting point. The minima are real and not wherever the optimiser happened to land.

## 5 What the data determine weakly, or not at all

This section is the point of the exercise, and is stated as plainly as the numbers allow.

**$n/p$  is separated by four points.** The statement needs care, because two different things are true of it.

The neutron channel is  $\langle X \rangle \propto X_u + 2X_d$  against the proton's  $2X_u + X_d$ , so the  $d$  block enters the neutron with twice the weight it has in the proton. Since  $\sigma_{TT} = \sum |T_{01}|^2 - \sum |U_{01}|^2$  and  $U_{01}$  is built from  $\bar{E}_T$ , the ratio  $n/p$  in  $\sigma_{TT}$  is essentially a measurement of  $\bar{E}_T^d$  relative to  $\bar{E}_T^u$ .

The fitted  $t$ -slopes of those two blocks differ by a factor eight,  $b_d = 5.37 \pm 0.36$  against  $b_u = 0.68 \pm 0.05$ , which invites the suspicion that this is the same flat valley that forced  $b_d = b_u$  in the  $H_T$  sector. It is not. Imposing the same tie on  $\bar{E}_T$  costs *2662 units of  $\chi^2$  for one parameter* —  $\chi^2/\text{ndf}$  from 1.39 to 3.13 — and the damage is spread over the whole database, not carried by the neutron: CLAS6  $\pi^0$  goes from 1.68 to 3.04 per point, CLAS6  $\eta$  from 1.04 to 5.05, eg1 from 1.85 to 12.38. For comparison, the same tie in the  $H_T$  sector costs two units, which is what a flat valley actually looks like. The steepness of  $\bar{E}_T^d$  is therefore a requirement of the data, not an artefact. (That test was started from the unconstrained solution and is an upper bound on the constrained minimum; given a factor 2.25 in  $\chi^2$  the conclusion does not depend on how tight the bound is.)

What the neutron supplies is not the steepness but the *separation*. In the proton combination  $2X_u + X_d$  the normalisation  $N_d$  and the slope  $b_d$  of the  $d$  block are degenerate —  $d$  sits at half the

weight of  $u$  and can trade one against the other. The neutron combination breaks that degeneracy, and it is the only thing in the database that does. Removing the four neutron points moves  $n/p$  from 1.06 to 2.15, and the movement is traceable almost entirely to  $b(\bar{E}_T^d)$  relaxing from 5.32 to 3.57; changing that one parameter alone already carries  $n/p$  to 1.45.

So  $n/p$  is constrained, but by four points with uncertainties of 42 to 90 %, and it is the least robust number in this report. During one day of otherwise minor and justified changes — correcting systematics, adding a normalisation nuisance, repairing a parameter bound — it moved through 1.00, 1.02, 0.98, 1.04, 1.06 while  $\chi^2/\text{ndf}$  changed in the third decimal. It should be quoted with that history attached, and a fifth neutron point would be worth more than a thousand proton points.

**One experiment fixes a phase.**  $\sigma_{T'} = 2 \text{Re}\langle T_{01}, U_{01} \rangle$  by eq. (4); the unpolarised data fix only the moduli. The  $A_{LL}$  moments of eg1-dvcs are the only observable in the database that sees this interference, and when eg1 is left out its own blind  $\chi^2$  is 41 per point, with  $\phi_w$  swinging from  $-0.70$  to  $+2.90$  — essentially a sign flip of  $\sigma_{T'}$ . Forty points from one experiment.

**$\sigma_{LT}$  is a systematic failure of the model.** It is the worst described quantity in three independent sets: 20.4 per point in the blind prediction of the CLAS12 cross sections, 4.7 in Hall A 2016, and the worst of the four observables in Hall A 2021. Since  $\sigma_{LT}$  carries the entire  $\cos\phi$  modulation of eq. (5), any event generator built on this model will inherit the failure in the shape of its  $\phi$  distribution.

**A single power of  $Q^2$  does not span the range.** Hall A 2016 and Hall A 2021 disagree with the model in opposite directions at the two ends: at  $Q^2 = 1.5$  the model undershoots  $\sigma_T$  ( $1616 \pm 145$  measured against 1135), at  $Q^2 = 8.3$  it overshoots  $\sigma_U$  by a factor 1.8. Hall A 2016 does not fit at all — 4.5 per point with its normalisation already dragged  $1.8\sigma$  — and these are the only Rosenbluth-separated proton points that exist. Admitting Hall A 2021 to the fit brings it from 13.1 to 2.34 per point, but the asymmetries pay: eg1 from 1.36 to 1.81, the CLAS12 beam-spin asymmetry from 1.04 to 1.53. The compromise is real and it is visible.

## 6 Comparison with Goloskokov–Kroll

Two independent comparisons put the GK prediction for  $\sigma_{LT'}$  far below the data. The published GK curve for the  $\eta$  beam-spin asymmetry (Zhao *et al.*, PLB **789** 426, figure 5) is near zero across all four bins where the data are 0.1 to 0.3. For  $\pi^0$ , a GK calculation over the 60 De Masi bins gives  $\alpha \approx 0.01\text{--}0.025$  against a published median of 0.080. The present model gives a median 0.077 on the same bins, but that is a fit, not a prediction; the fair statement is that when the De Masi  $\phi$  distributions are removed from the fit entirely they are still predicted at 1.02 per point, so the magnitude of  $\sigma_{LT'}$  demanded by all the *other* data is about five times what GK produces.

Throughout,  $\alpha$  is the amplitude of  $A_{LU} = \alpha \sin\phi / (1 + \beta \cos\phi + \gamma \cos 2\phi)$  with  $A_{LU}$  already divided by the beam polarisation, so  $\alpha = \sqrt{2\epsilon(1-\epsilon)} \sigma_{LT'} / (\sigma_T + \epsilon\sigma_L)$  and not the bare ratio.

## 7 Figures

Every figure shows the data of one set against the fit. A title reading NOT IN THE FIT marks a blind prediction. Data are drawn in red, the model in blue. Each panel carries its own kinematics and  $\chi^2$ ; the model curve includes the fitted normalisation of table 2 where the set has one, so the curve is what the fit actually compares against the points.

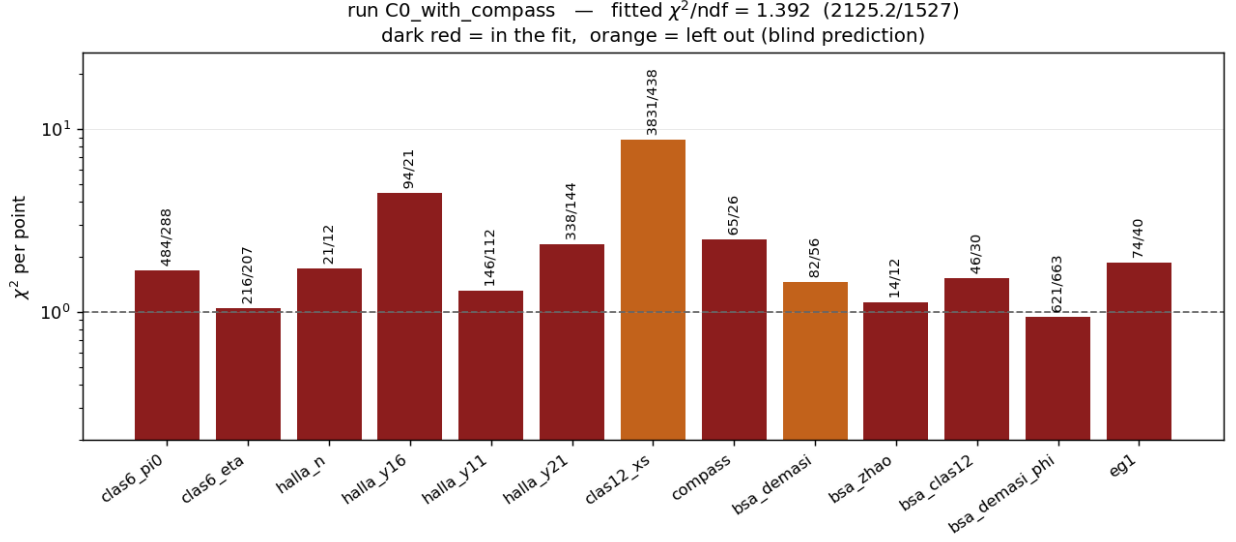


Figure 1: All thirteen sets at a glance:  $\chi^2$  per point, fitted and blind.

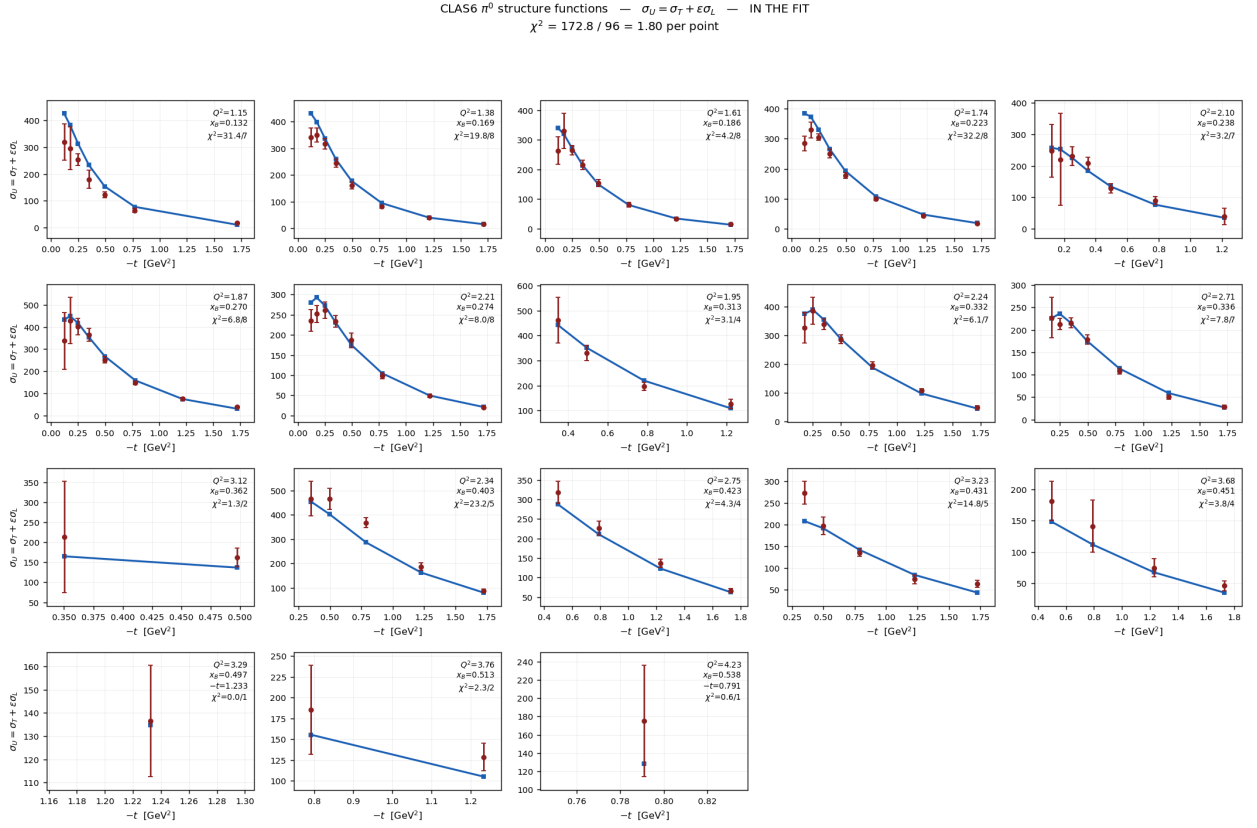


Figure 2: CLAS6  $\pi^0$ ,  $\sigma_U = \sigma_T + \epsilon\sigma_L$ , with re-computed radiative corrections.



CLAS6  $\pi^0$  structure functions —  $\sigma_{LT}$  — IN THE FIT  
 $\chi^2 = 191.0 / 96 = 1.99$  per point

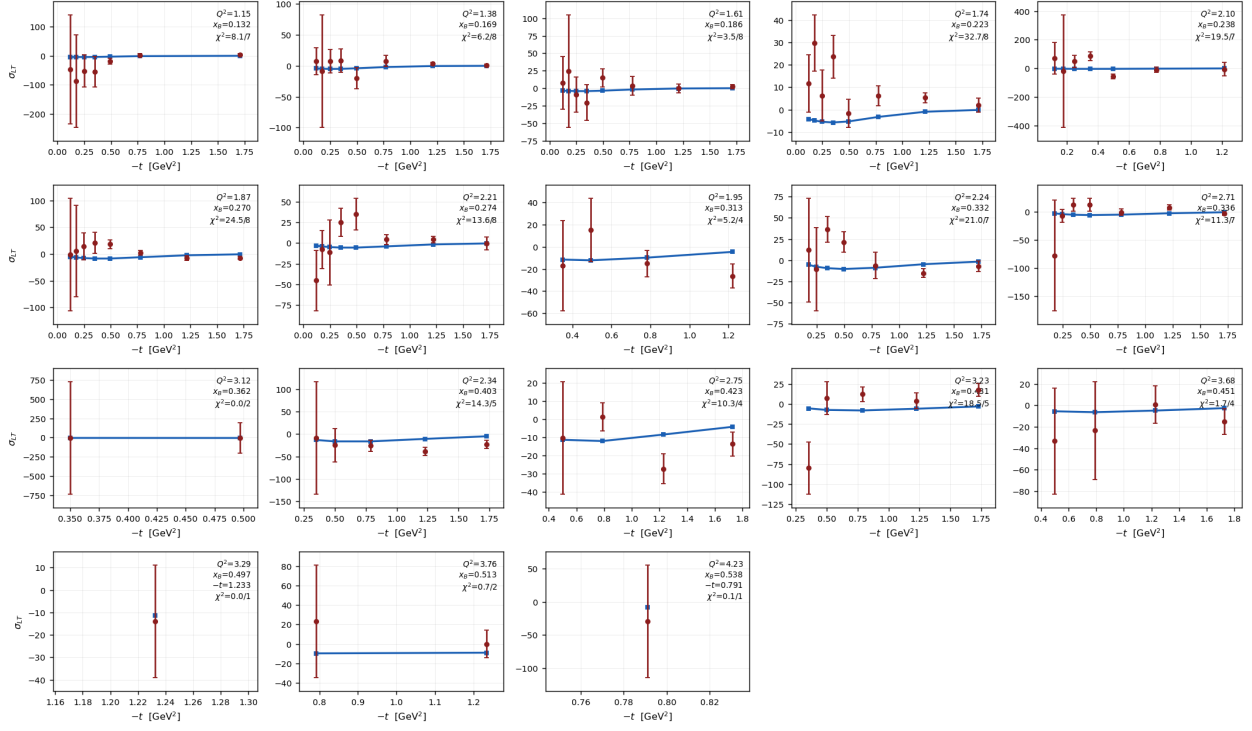


Figure 3: CLAS6  $\pi^0$ ,  $\sigma_{LT}$ .

CLAS6  $\pi^0$  structure functions —  $\sigma_{TT}$  — IN THE FIT  
 $\chi^2 = 120.4 / 96 = 1.25$  per point

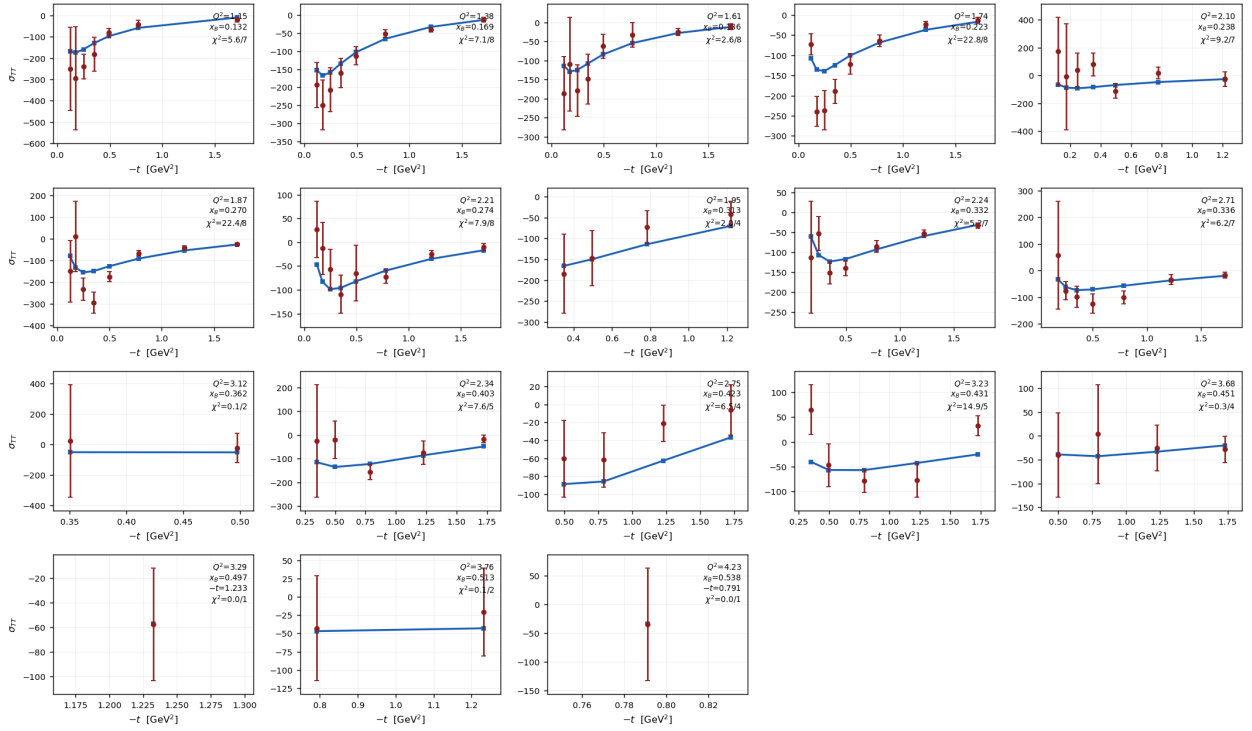


Figure 4: CLAS6  $\pi^0$ ,  $\sigma_{TT}$ .

CLAS6  $\eta$  structure functions —  $\sigma_U = \sigma_T + \epsilon\sigma_L$  — IN THE FIT  
 $\chi^2 = 100.4 / 69 = 1.46$  per point

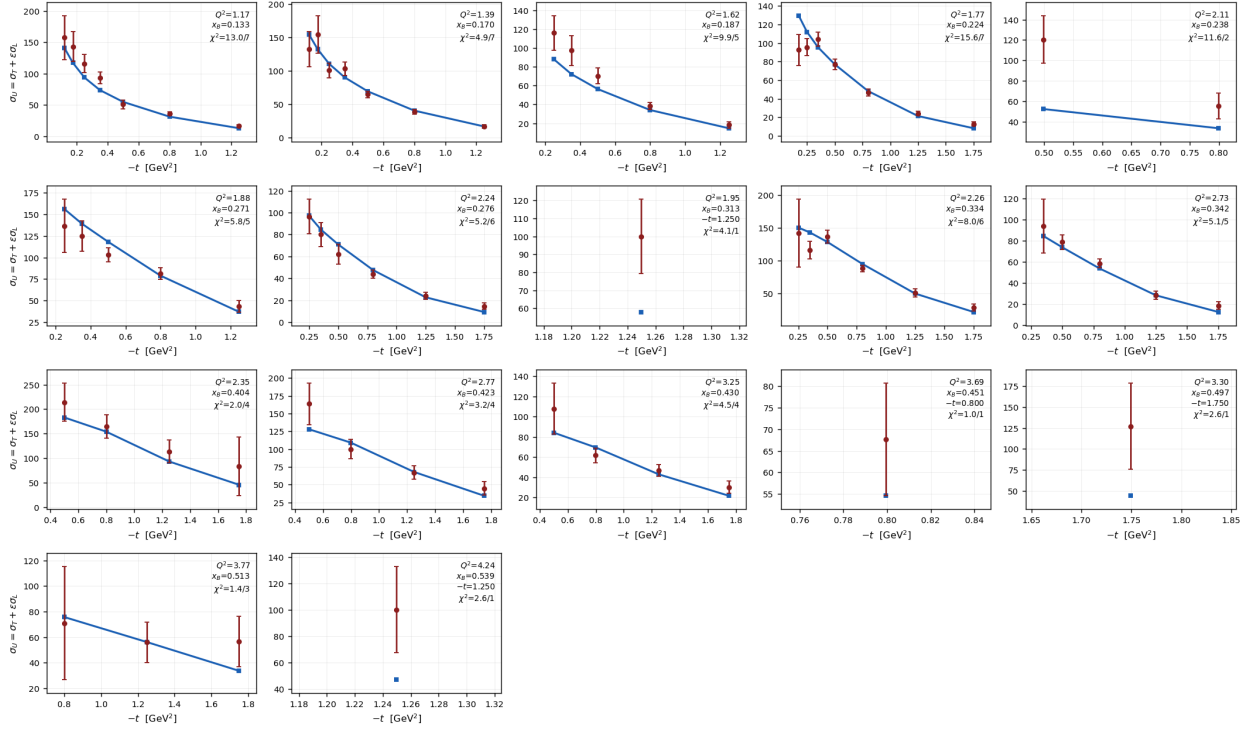


Figure 5: CLAS6  $\eta$ ,  $\sigma_U$ . This channel probes  $2X_u - X_d$ .

CLAS6  $\eta$  structure functions —  $\sigma_{LT}$  — IN THE FIT  
 $\chi^2 = 46.3 / 69 = 0.67$  per point

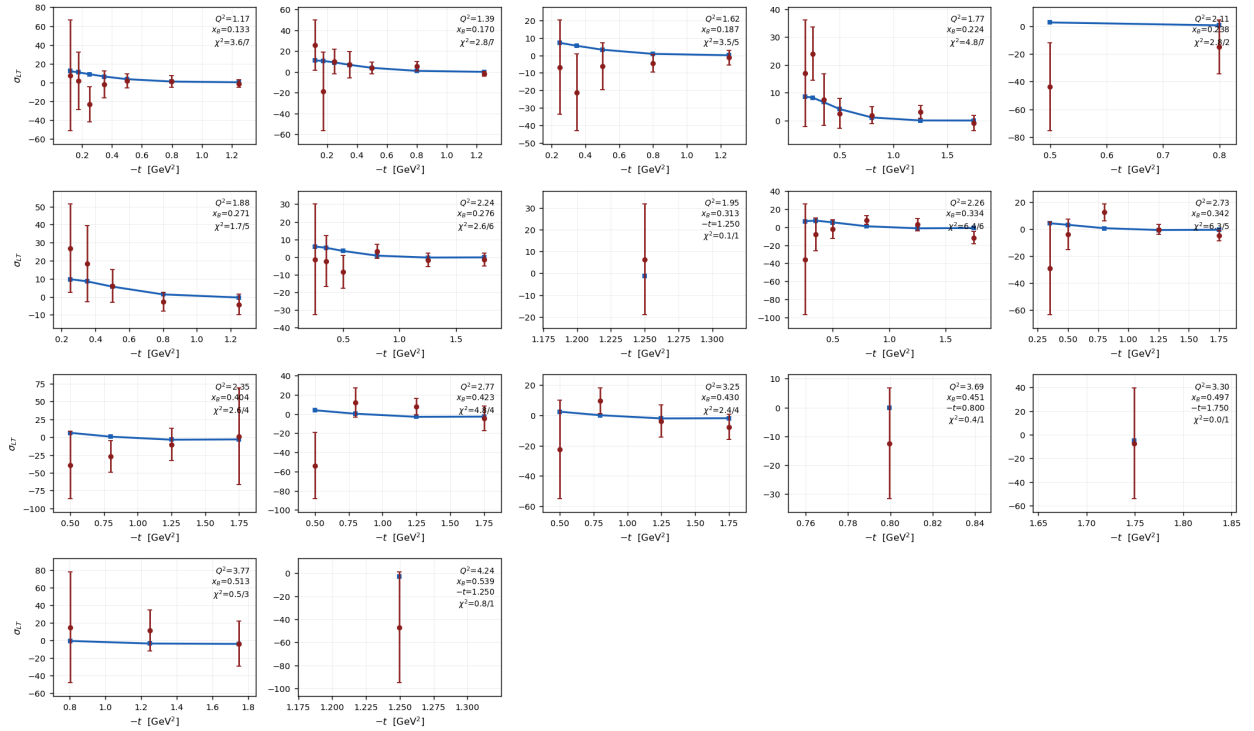


Figure 6: CLAS6  $\eta$ ,  $\sigma_{LT}$ .

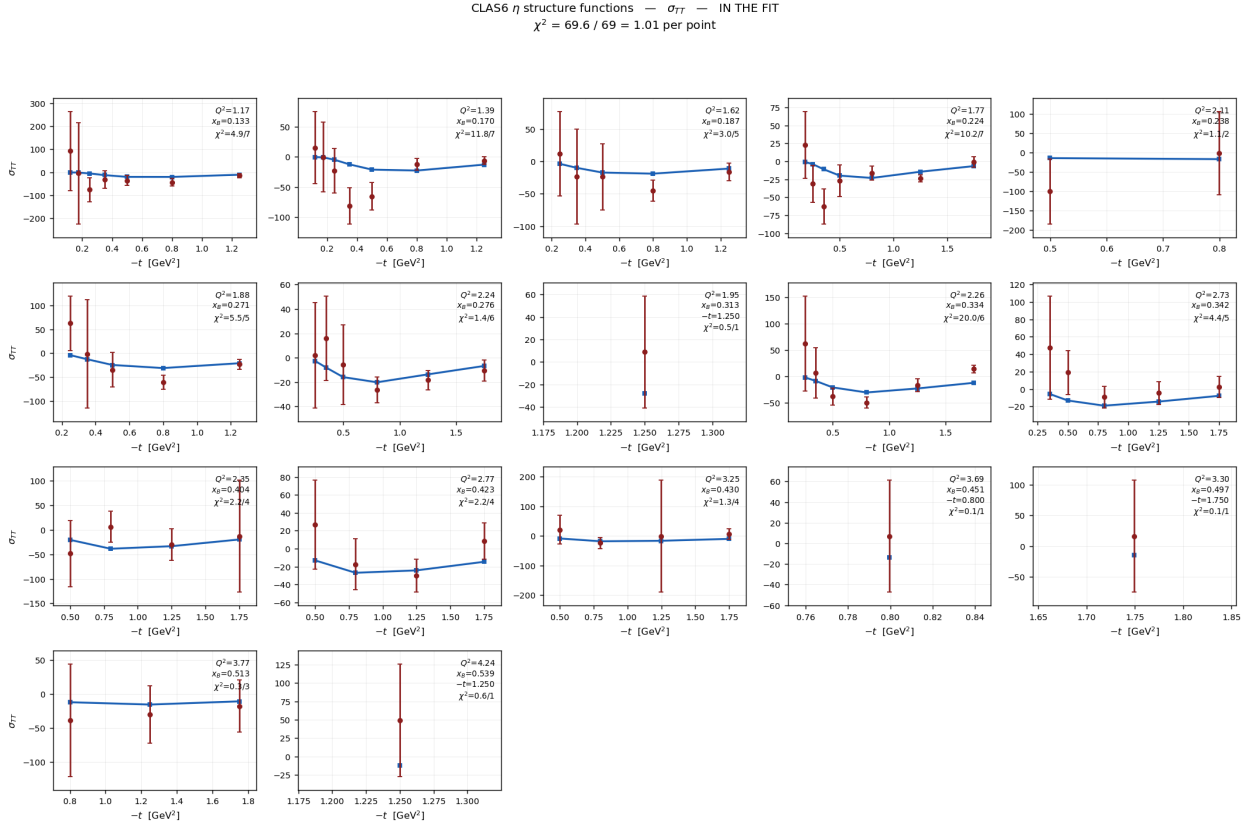


Figure 7: CLAS6  $\eta$ ,  $\sigma_{TT}$ .

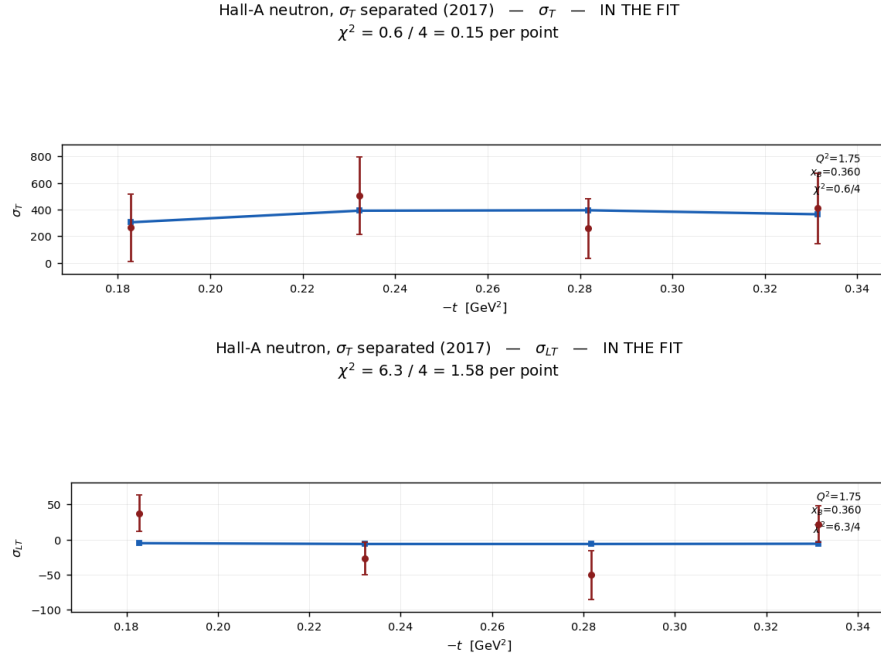


Figure 8: Hall A 2017,  $\pi^0$  off the *neutron*:  $\sigma_T$  and  $\sigma_{LT}$ . Four points, uncertainties of 42 to 90 %, and the sole constraint on  $n/p$ .

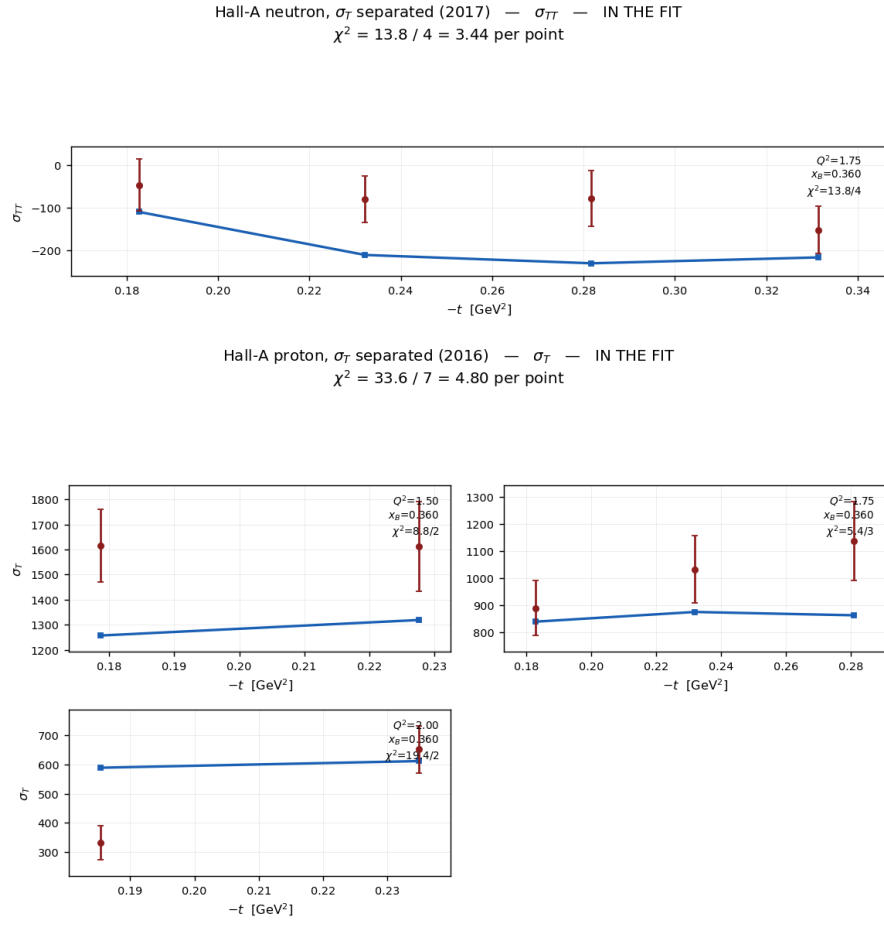
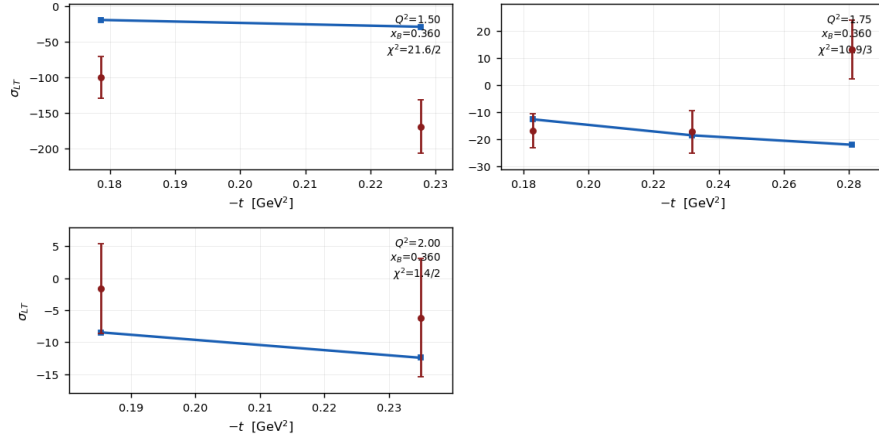


Figure 9: Neutron  $\sigma_{TT}$  (top) and Hall A 2016 proton  $\sigma_T$  (bottom). The latter is the worst-described set in the fit at 4.5 per point, with its normalisation already pulled  $1.8\sigma$ .

Hall-A proton,  $\sigma_T$  separated (2016) —  $\sigma_{LT}$  — IN THE FIT  
 $\chi^2 = 33.9 / 7 = 4.84$  per point



Hall-A proton,  $\sigma_T$  separated (2016) —  $\sigma_{TT}$  — IN THE FIT  
 $\chi^2 = 26.5 / 7 = 3.78$  per point

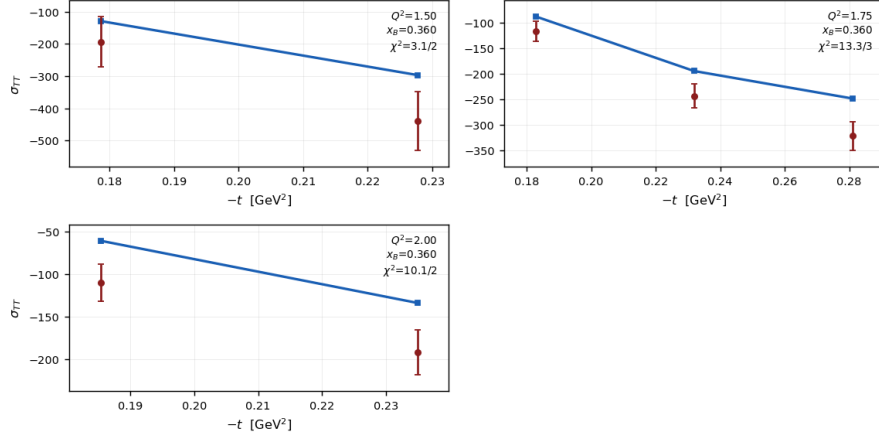


Figure 10: Hall A 2016,  $\sigma_{LT}$  and  $\sigma_{TT}$ .

Hall-A proton 6 GeV (2011) —  $\sigma_U = \sigma_T + \varepsilon\sigma_L$  — IN THE FIT  
 $\chi^2 = 46.1 / 28 = 1.65$  per point

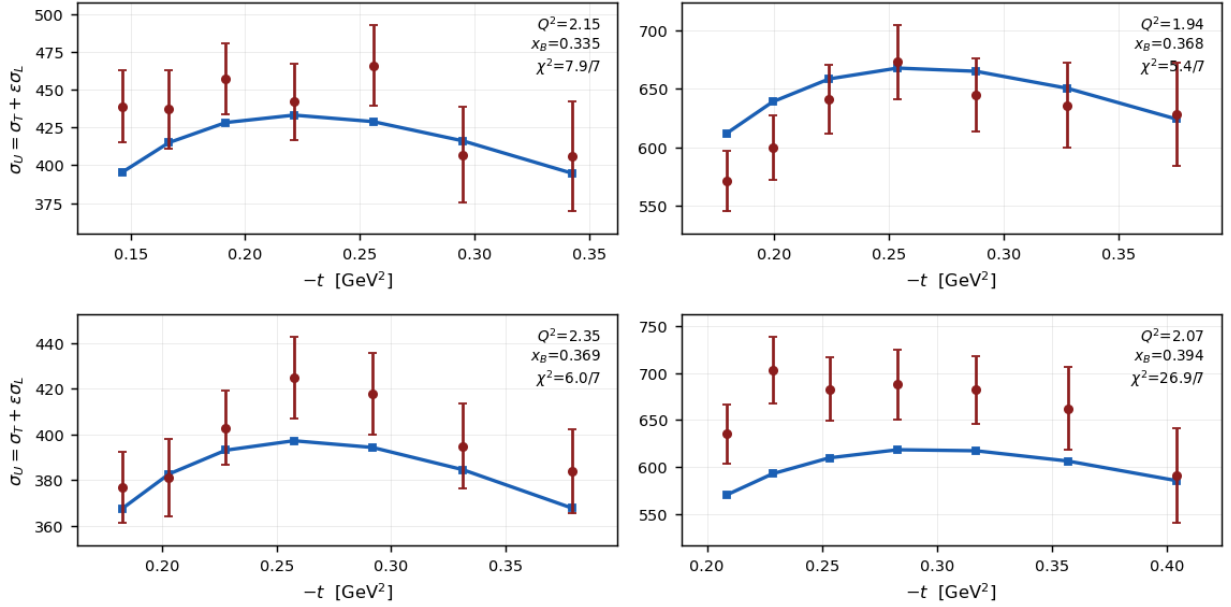
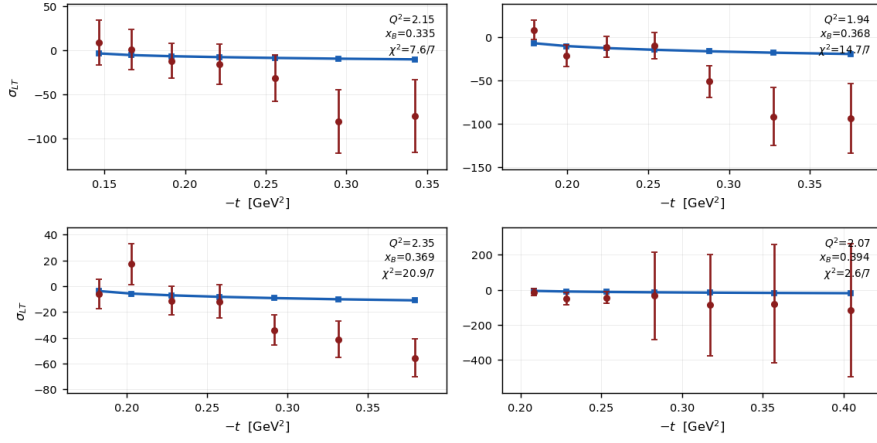


Figure 11: Hall A 2011,  $\sigma_U$ . This set is published in the Drechsel–Tiator normalisation;  $\sigma_{LT}$  and  $\sigma_{LT'}$  are converted by  $\gamma = 2Mx_B/\sqrt{Q^2}$  before comparison.

Hall-A proton 6 GeV (2011) —  $\sigma_{LT}$  — IN THE FIT  
 $\chi^2 = 45.8 / 28 = 1.63$  per point



Hall-A proton 6 GeV (2011) —  $\sigma_{TT}$  — IN THE FIT  
 $\chi^2 = 38.4 / 28 = 1.37$  per point

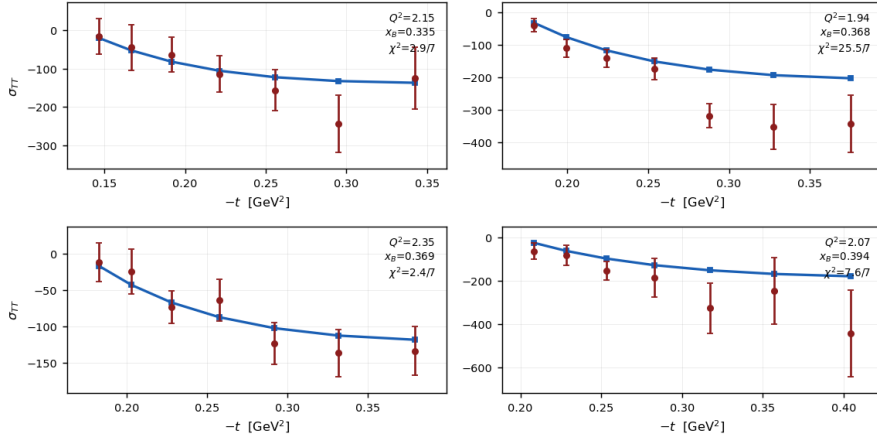


Figure 12: Hall A 2011,  $\sigma_{LT}$  and  $\sigma_{TT}$ .

Hall-A proton 6 GeV (2011) —  $\sigma_{LT'}$  — IN THE FIT  
 $\chi^2 = 16.0 / 28 = 0.57$  per point

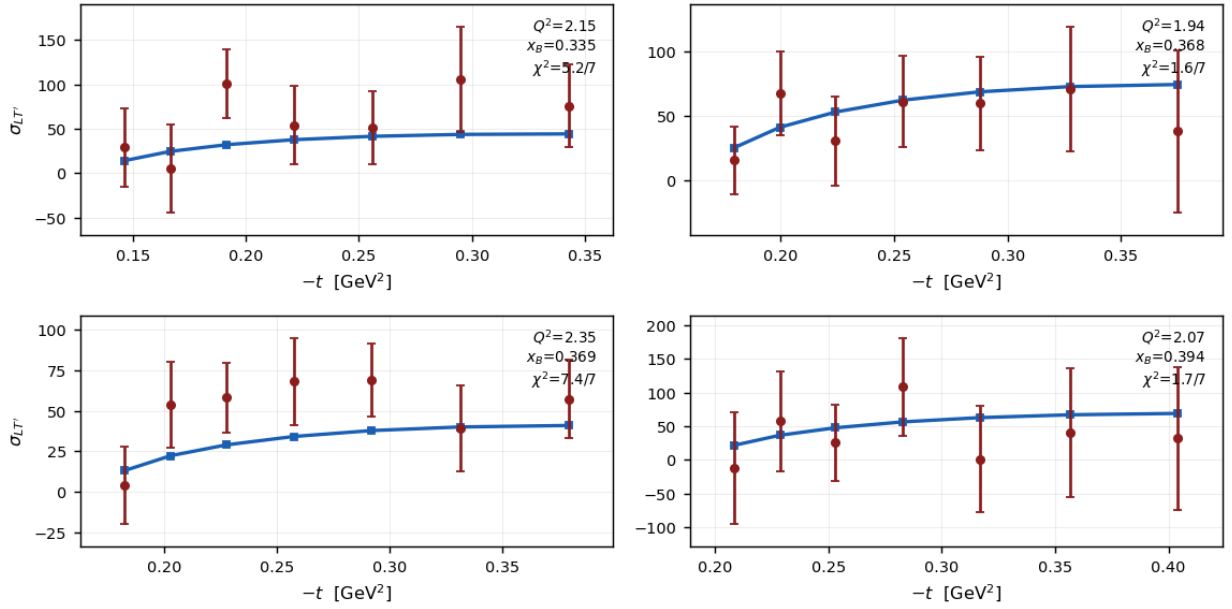


Figure 13: Hall A 2011,  $\sigma_{LT'}$ .



Hall-A proton 12 GeV (2021) —  $\sigma_U = \sigma_T + \varepsilon \sigma_L$  — IN THE FIT  
 $\chi^2 = 103.9 / 36 = 2.89$  per point

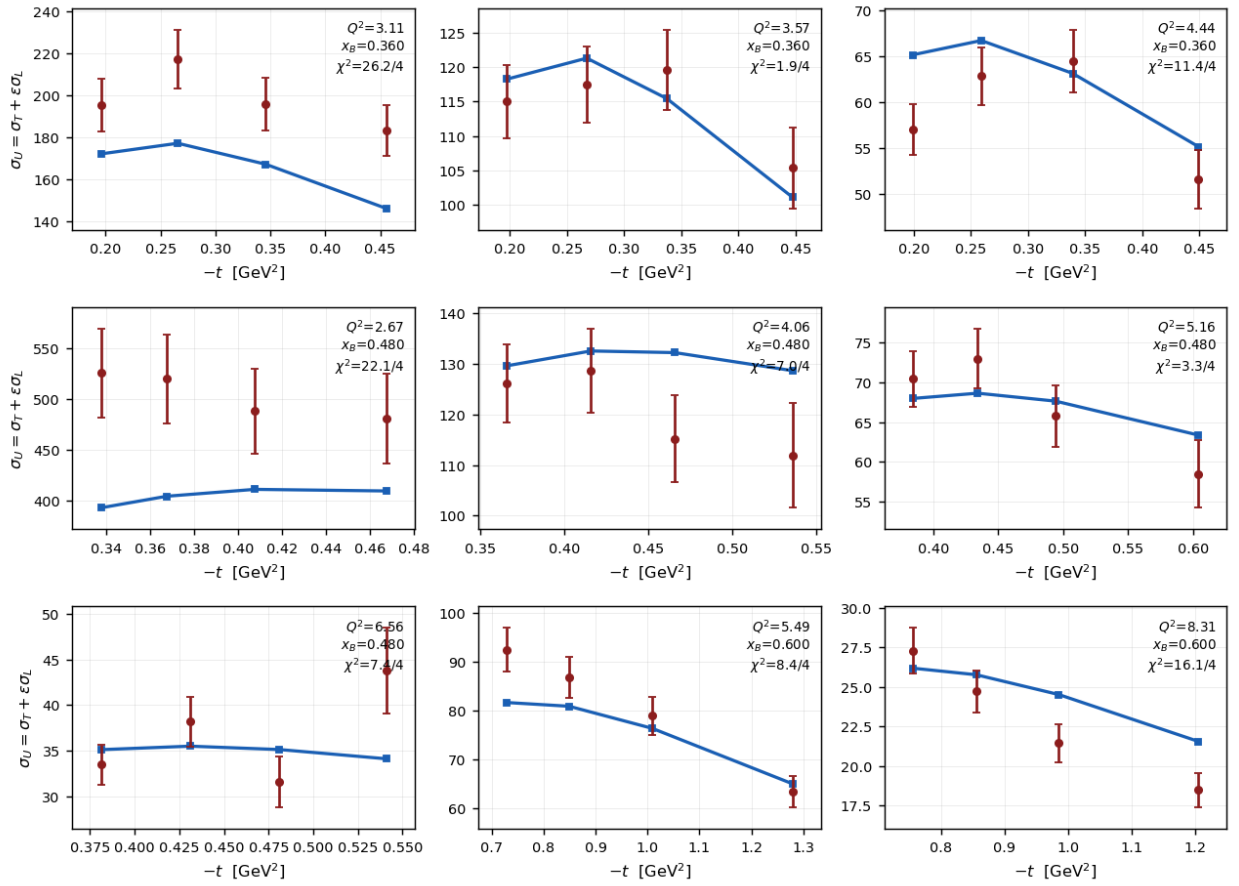
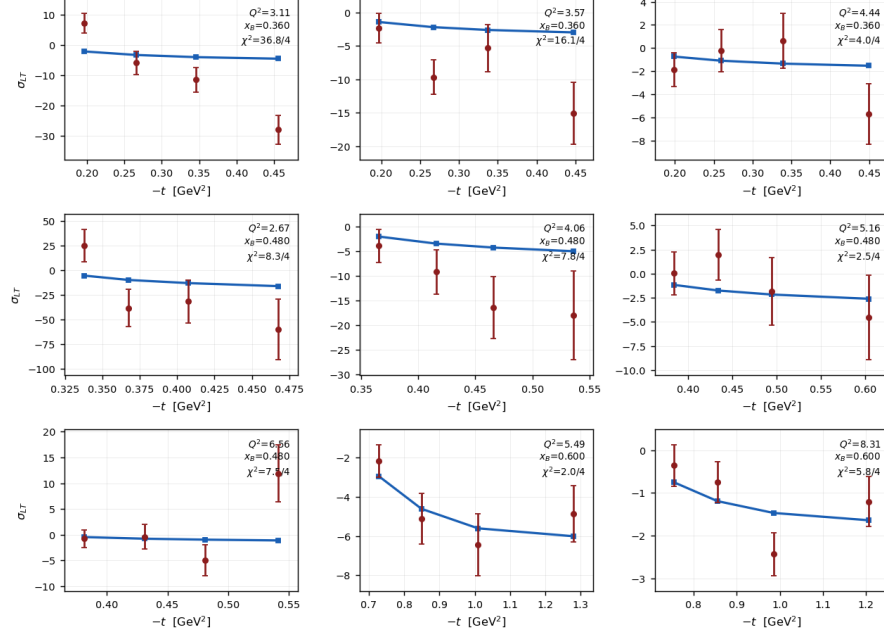


Figure 14: Hall A 2021,  $\sigma_U$ . The only data above  $Q^2 = 4.3$ , reaching 8.31 GeV<sup>2</sup>; left out of the fit the model overshoots these points by a factor 1.8.

Hall-A proton 12 GeV (2021) —  $\sigma_{LT}$  — IN THE FIT  
 $\chi^2 = 90.8 / 36 = 2.52$  per point



Hall-A proton 12 GeV (2021) —  $\sigma_{TT}$  — IN THE FIT  
 $\chi^2 = 90.7 / 36 = 2.52$  per point

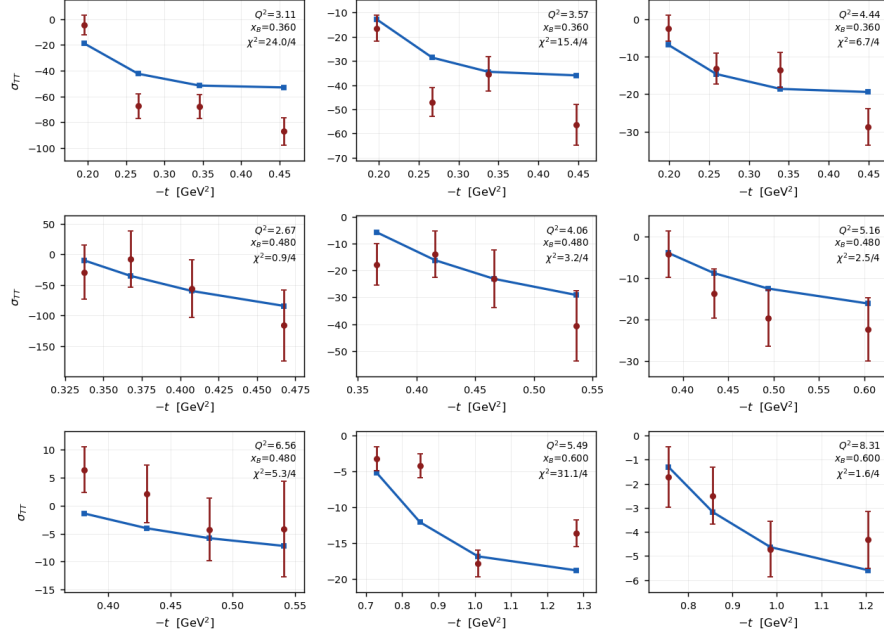


Figure 15: Hall A 2021,  $\sigma_{LT}$  and  $\sigma_{TT}$ .

Hall-A proton 12 GeV (2021) —  $\sigma_{LT'}$  — IN THE FIT  
 $\chi^2 = 52.8 / 36 = 1.47$  per point

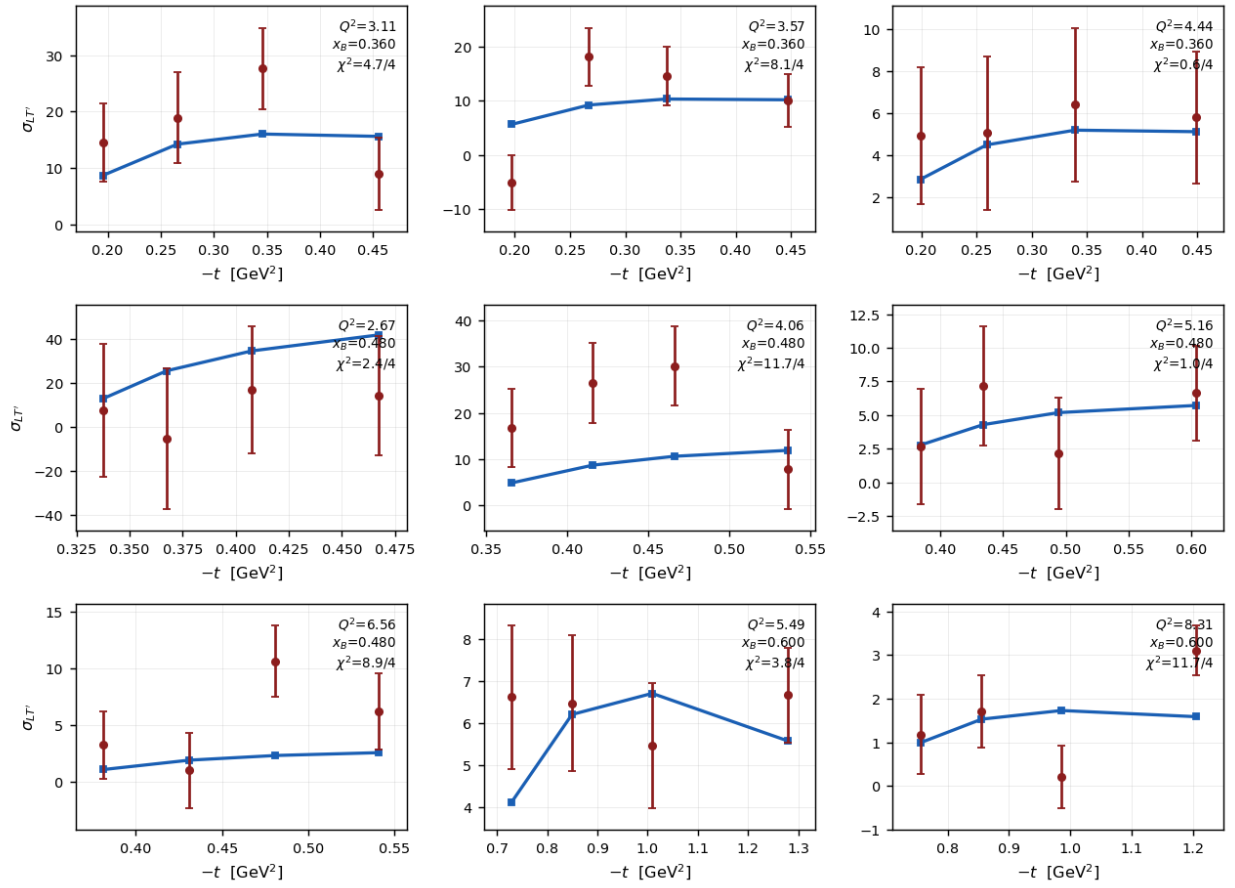


Figure 16: Hall A 2021,  $\sigma_{LT'}$ .

CLAS12 cross sections, preliminary —  $\sigma_U = \sigma_U + \epsilon \sigma_U$  — NOT in the fit (blind prediction)  
 $\chi^2 = 347.4 / 146 = 2.38$  per point

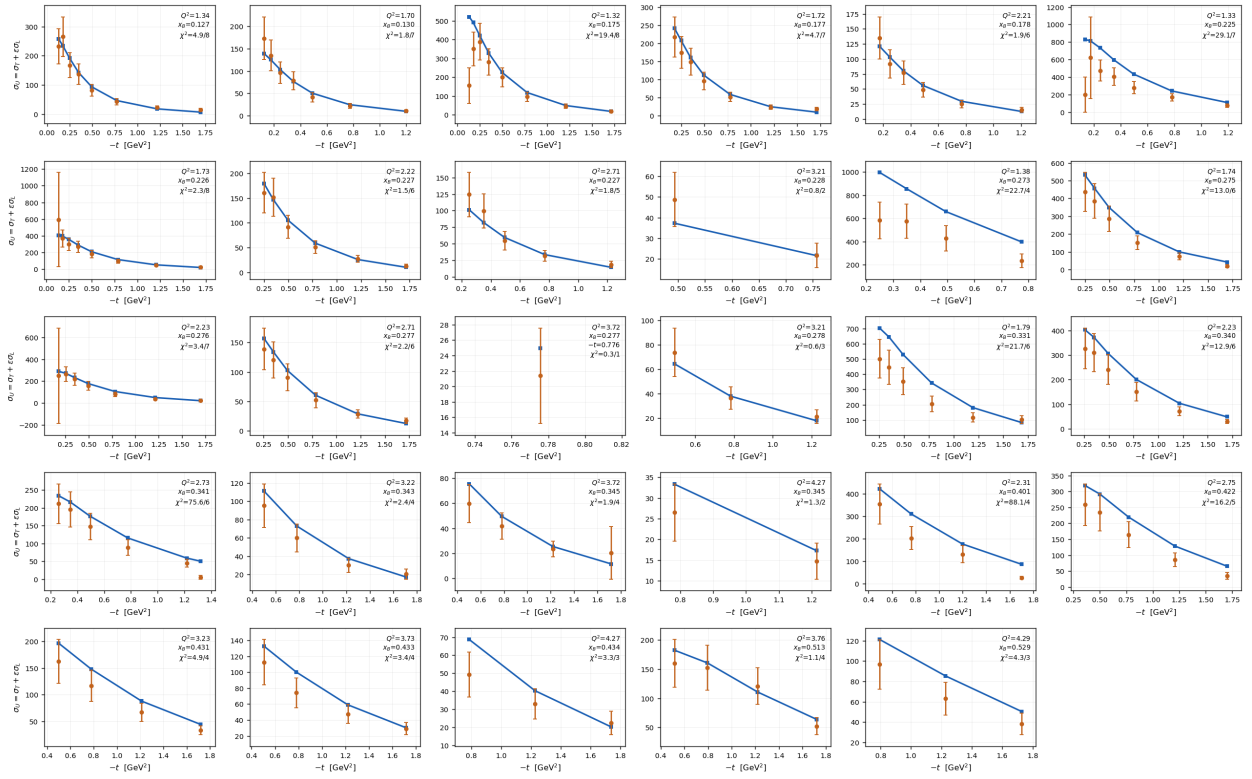


Figure 17: CLAS12 cross sections,  $\sigma_U$  — *not* in the fit, and preliminary. Predicted at 2.39 per point.

CLAS12 cross sections, preliminary  $\sigma_T$  — NOT in the fit (blind prediction)  
 $\chi^2 = 2953.4 / 146 = 20.23$  per point

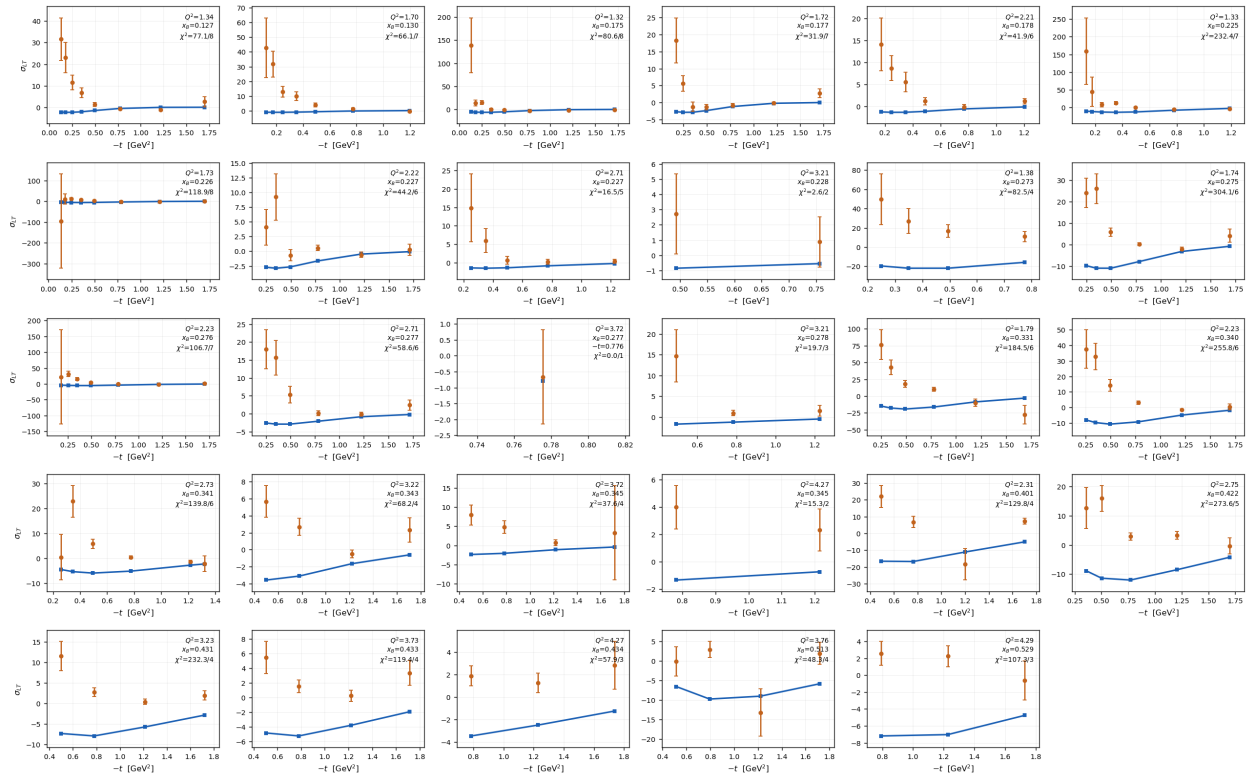


Figure 18: CLAS12,  $\sigma_{LT}$ , blind: 20.4 per point. This is the clearest single picture of the model's failure in  $\sigma_{LT}$ .

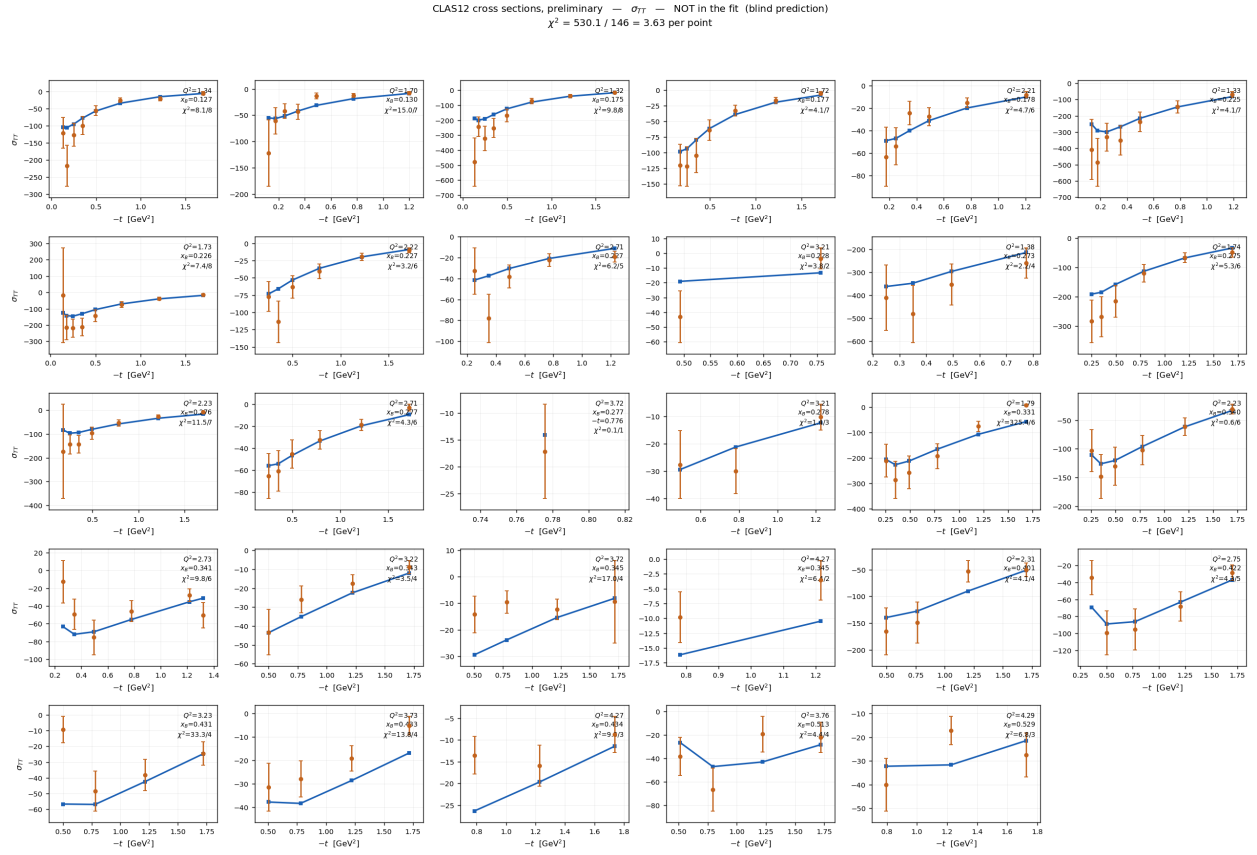


Figure 19: CLAS12,  $\sigma_{TT}$ , blind: 3.51 per point.

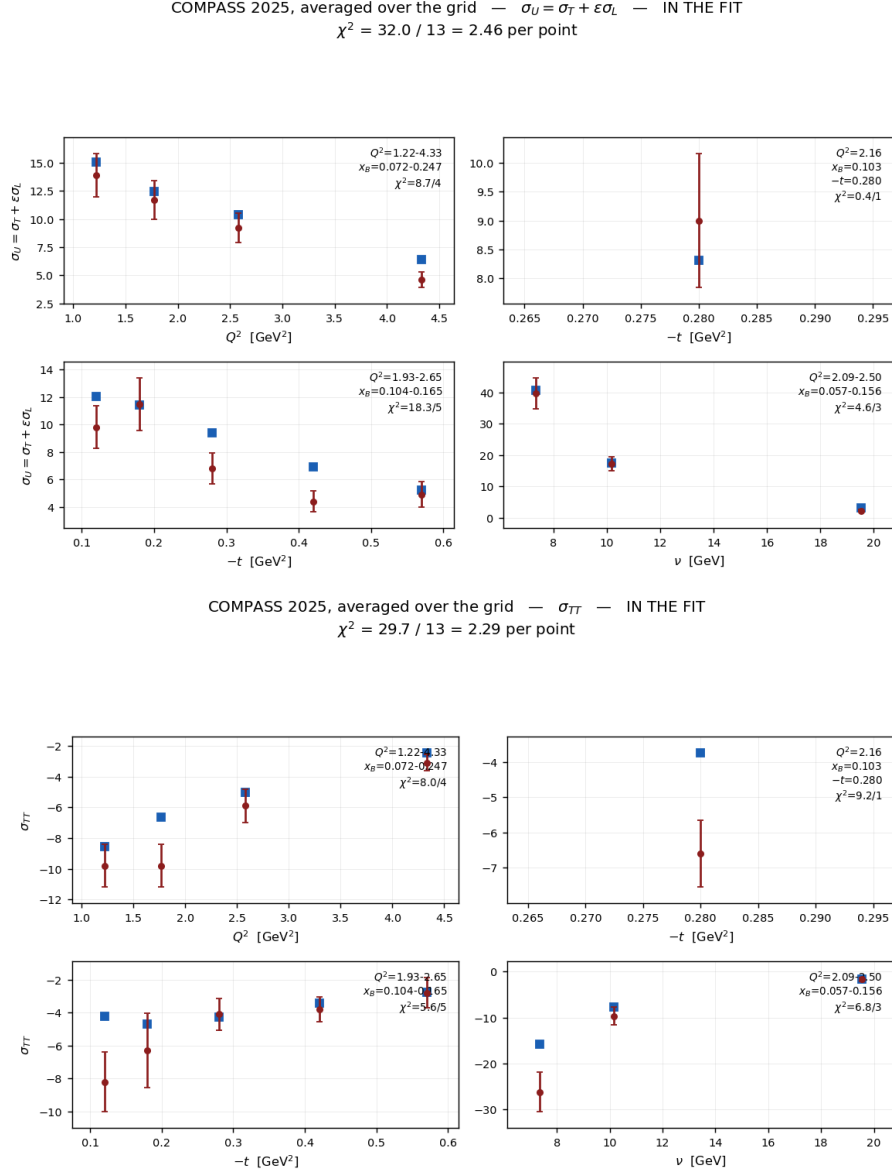


Figure 20: COMPASS 2025. The model is averaged over the four-dimensional grid of the measurement, eq. (16) of the paper, weighted by cell volume: a cell rectangular in  $(Q^2, \nu)$  is not rectangular in  $x_B$ , and evaluating at the quoted mean kinematics overshoots by a factor five to eight.

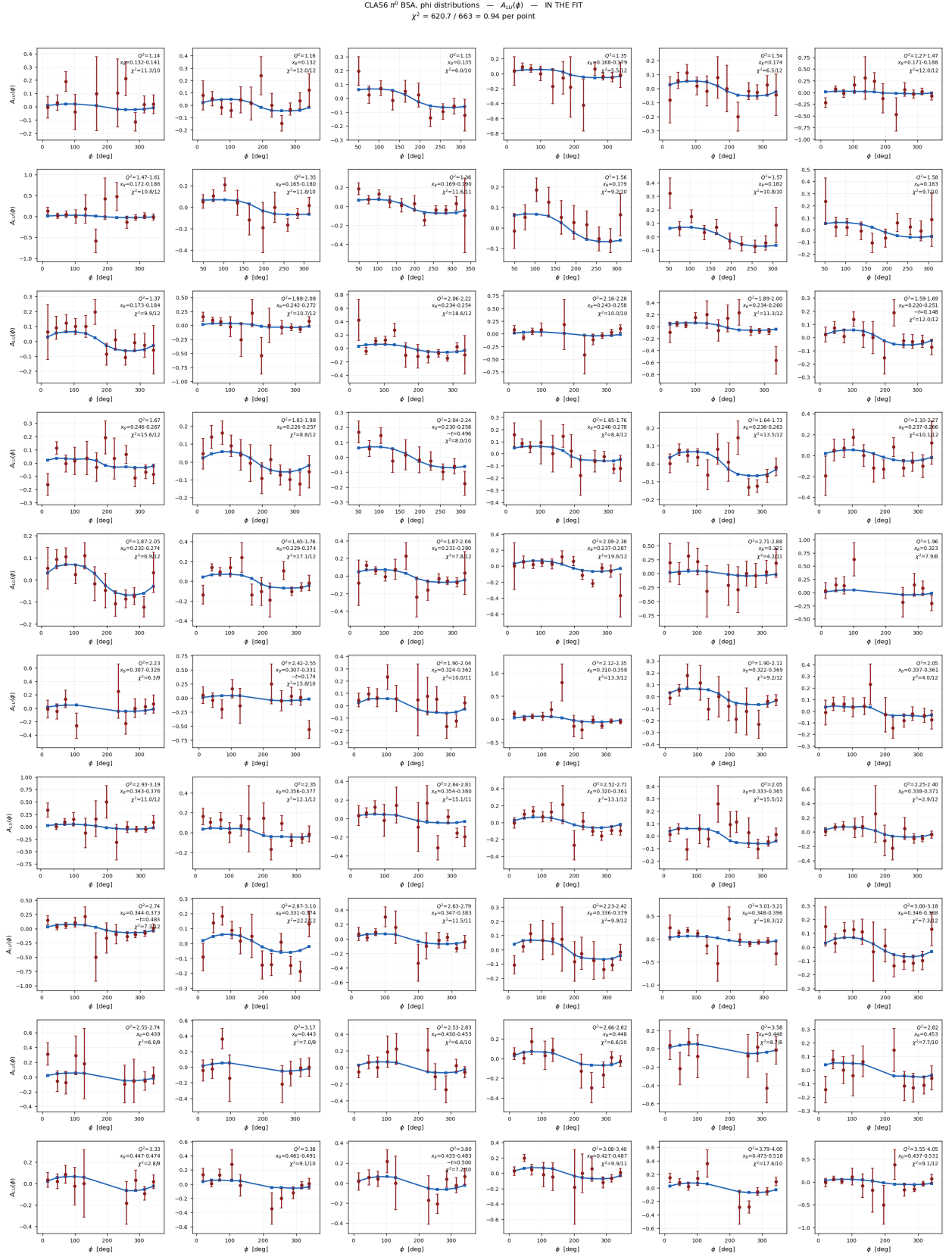


Figure 21: De Masi  $A_{LU}(\phi)$ : all 60 bins, 12 points in  $\phi$  each. The model supplies the denominator of eq. (5) from its own  $\sigma_{LT}$  and  $\sigma_{TT}$ , so no free  $\beta, \gamma$  are fitted per bin — a three-parameter fit per bin is unstable, the denominator passing near zero in 13 of the 60.



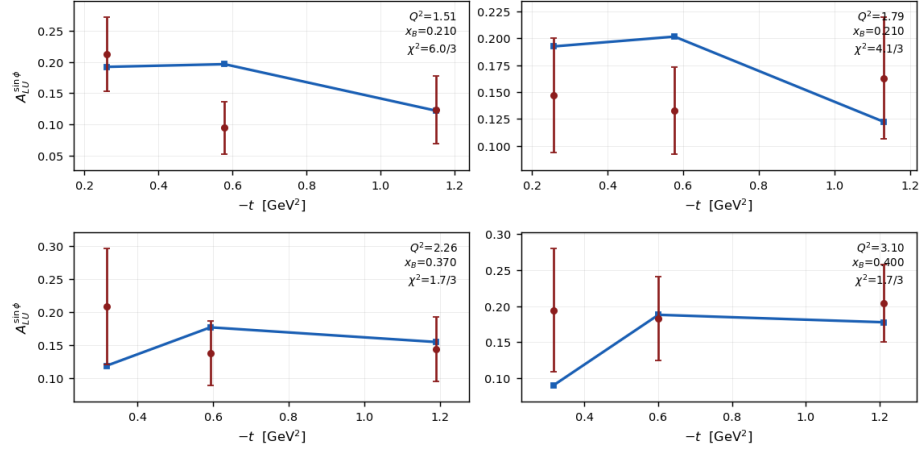
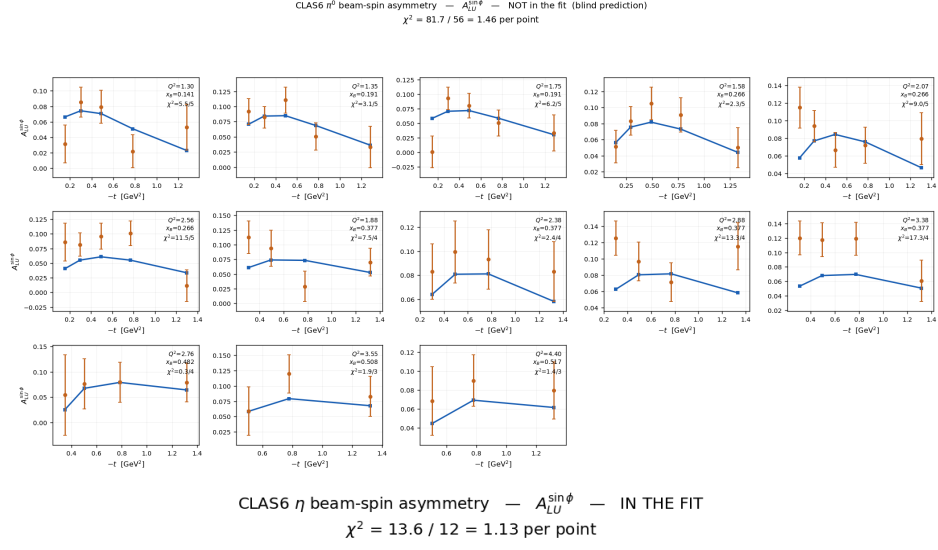


Figure 22: Top: the published De Masi  $\alpha$ , kept out of the fit as the same measurement as the figure above. Bottom: the Zhao  $\eta$  beam-spin asymmetry, where the published GK curve is near zero.

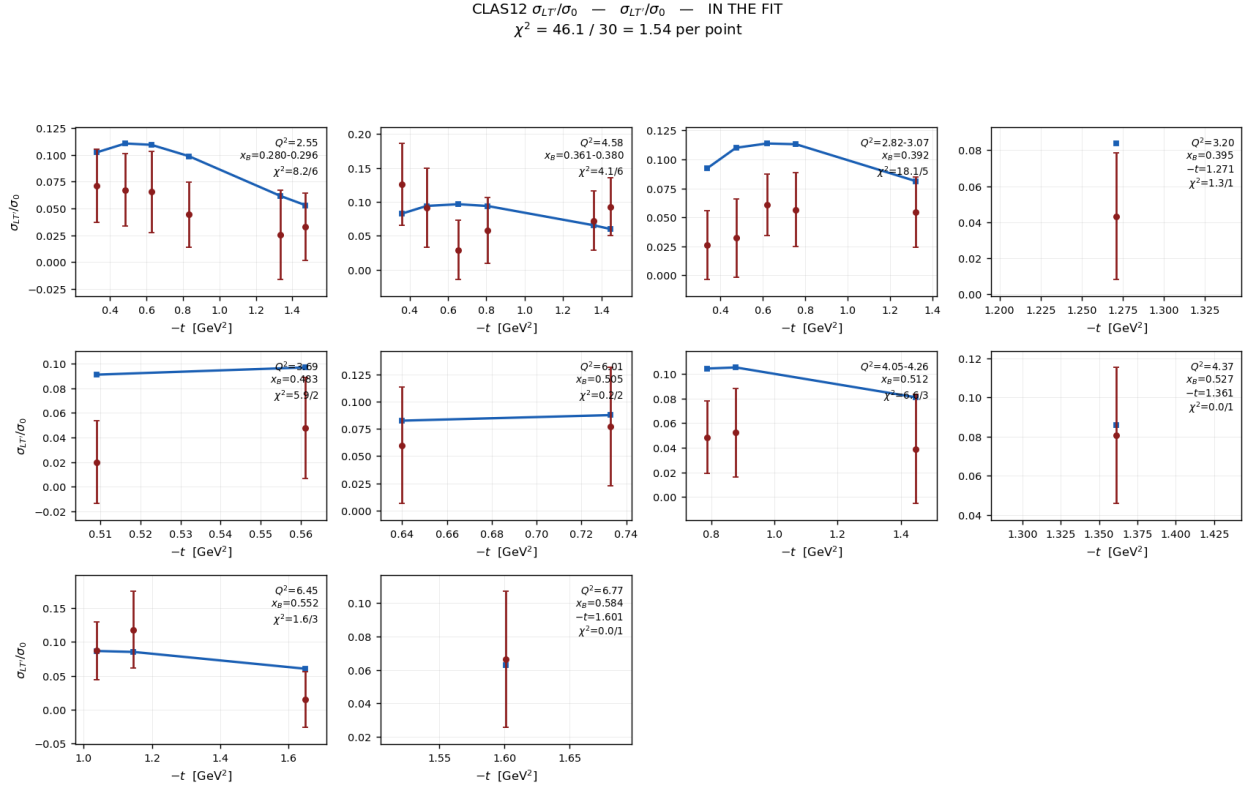


Figure 23: CLAS12  $\sigma_{LT'}/\sigma_0$ .

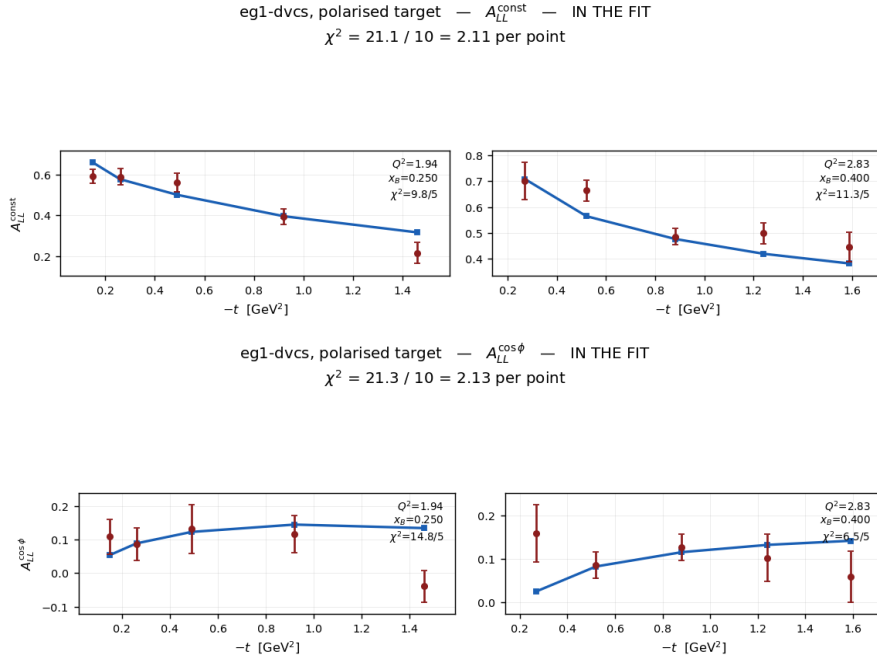
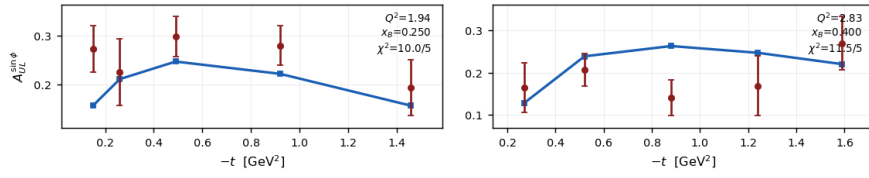


Figure 24: eg1-dvcs, polarised target: the  $A_{LL}$  moments, at the experiment's two settings. These are the only observables in the database sensitive to the relative phase of  $T_{01}$  and  $U_{01}$ ; without them it is unconstrained.

eg1-dvcs, polarised target —  $A_{UL}^{\sin\phi}$  — IN THE FIT  
 $\chi^2 = 21.4 / 10 = 2.14$  per point



eg1-dvcs, polarised target —  $A_{UL}^{\sin 2\phi}$  — IN THE FIT  
 $\chi^2 = 10.1 / 10 = 1.01$  per point

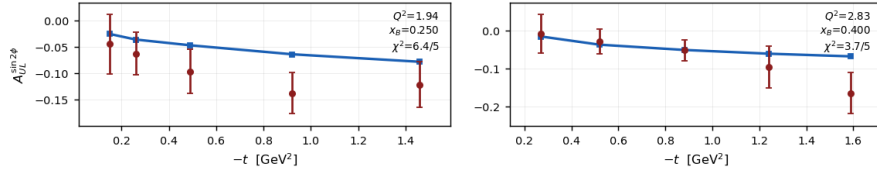


Figure 25: eg1-dvcs, the  $A_{UL}$  moments.

## Reproducing this

Code, data and the full history are at [https://github.com/vkubarovsky/rc\\_iter1](https://github.com/vkubarovsky/rc_iter1). The fit shown is run `C0_with_compass`; its parameter vector is `runs/C0_with_compass/fitpar.npy` and every number in the tables above is generated from it by `tex/make_tables.py`, so nothing here is transcribed by hand. The data are `pi0_eta_database.xlsx`, md5 `04c3adf9d6e6b8d2b1b3729f4ff79d1d`; `export_db.py --check` verifies that the CSV cache the fit reads still matches the workbook.